Metrological Characterisation of a Low-Cost Ultrasonic Indoor Positioning System on a Mobile Robot and Correction of a Beacon Configuration Error

[TODO: working title]

Andrea D'Antona
Department of Engineering
University of Ferrara
44122 Ferrara, Italy
andrea.dantona@unife.it

Berat Baran Balci
[TODO: department]
[TODO: university]
[TODO: city, country]
[TODO: e-mail]

Marcello Bonfè
Department of Engineering
University of Ferrara
44122 Ferrara, Italy
marcello.bonfe@unife.it

***Abstract*—[TODO: 150–200 words, write last. Problem: low-cost ultrasonic indoor positioning on a mobile robot, reference only in part of the area. What was done: Marvelmind on a KELO robot against OptiTrack in two campaigns at the same robot poses (rotations, triangles at two speeds, rests; out-and-back test). Main numbers: precision $1.6\,\mathrm{mm}$ (Allan, $1\,\mathrm{s}$) vs accuracy 3–7 cm at rest; about 11 cm RMS while moving; comparison uncertainty dominated by the registration (5 cm). A wrong height of one beacon raised the p95 error up to 1.4 m; discarding inconsistent samples is not enough; recomputing the recorded positions with the correct map, without the reference, made the first campaign agree with the second one, recorded with the correct configuration.]**

***Index Terms*—indoor positioning, ultrasonic positioning, motion capture, measurement uncertainty, mobile robots**

I. Introduction

[TODO: Low-cost indoor positioning systems (IPS) for mobile robots; their accuracy must be measured on the robot. Motion capture gives sub-millimetre reference positions, but only in a small volume.]

[TODO: Configuration of ultrasonic IPS (beacon positions and heights) is done by hand; a single wrong parameter is easy to miss and degrades the measurement. Can the data be saved afterwards?]

Contributions:

- [TODO: characterisation of Marvelmind on a mobile robot against OptiTrack: at rest (precision vs accuracy, repeated visits, heading), while moving, outside the reference area;]
- [TODO: effect of a wrong beacon height on the measurement;]
- [TODO: a method to recompute the recorded positions with the corrected map, without the reference, using the known height of the mobile beacon as a check, validated by a second campaign at the same robot poses with the correct configuration.]

II. Related Work

[TODO: Indoor positioning technologies and surveys [1], [2].] [TODO: Published evaluations of Marvelmind and other ultrasonic IPS: search IEEE Xplore and add.] [TODO: Motion capture as reference and its own accuracy [3].] [TODO: Trajectory error metrics (ATE) [4]; uncertainty evaluation [5]; Allan deviation [6].]

III. Experimental Setup

A. Marvelmind

[TODO: Four stationary beacons (B1, B2, B4, B5) at 2.40–2.60 m, Super-Beacon hardware [TODO: check], firmware 8.42, 3D navigation, update rate $5.8\,\mathrm{Hz}$, filters off, air temperature setting $23\,^{\circ}\mathrm{C}$. Hedgehog on the robot, read through the modem (binary protocol, device time stamp). Fig. 1(a).]

B. Reference: OptiTrack

[TODO: V120:Trio, $120\,\mathrm{Hz}$, Motive 3.5; covered area about $2.5\,\mathrm{m} \times 2.5\,\mathrm{m}$ in a corner of the beacon area; rigid body with 13 markers; static noise below $0.1\,\mathrm{mm}$, mean marker error about $1\,\mathrm{mm}$.]

C. Robot and mounting

[TODO: KELO driven with a joystick. Two hedgehog mountings; lever arm with respect to the rigid body pivot $(-0.144, -0.141)\,\mathrm{m}$ and $(0.214, 0.241)\,\mathrm{m}$, estimated from the data ($52\,\mathrm{cm}$ apart; tape: about $45\,\mathrm{cm}$). Rigid body: two groups of 5 markers on the mast and 3 markers $18.5\,\mathrm{cm}$ lower, at the height of the hedgehog ($0.69\,\mathrm{m}$ from the floor, $0.70\,\mathrm{m}$ reported by Marvelmind). Fig. 1(b).]

D. Logging and synchronisation

[TODO: Both systems on the same PC (`record.py`); Marvelmind device clock drift corrected; residual delay $0.105\,\mathrm{s}$ (Section IV-B).]

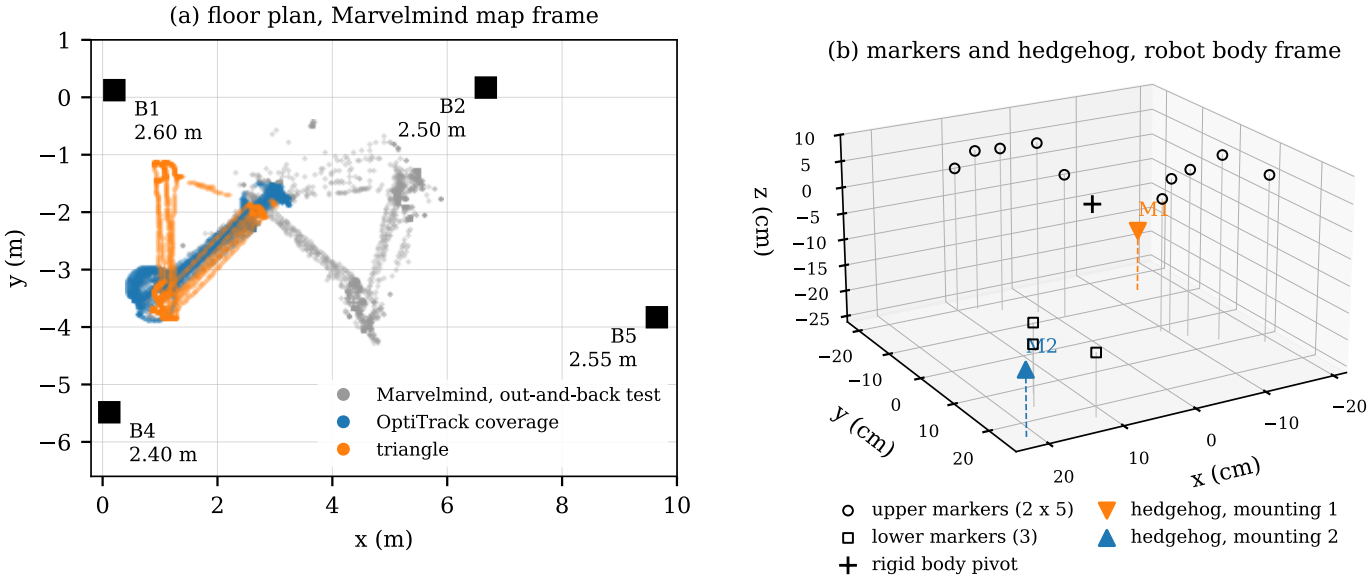


Fig. 1. Experimental setup. (a) Floor plan with the stationary beacons (height above the floor), the area covered by OptiTrack, the triangle path and the out-and-back path. (b) OptiTrack markers on the robot, rigid body pivot and position of the Marvelmind hedgehog in the two mountings (robot body frame). [TODO: add the photo of the robot]

TABLE I
TEST CAMPAIGNS. THE ROBOT POSES (TRIANGLE VERTICES AND HEADINGS) ARE THE SAME IN BOTH.

Test	Campaign 1	Campaign 2
Rotation in place, 4 headings	2 runs	2 runs
Triangle, 3 vertices	fast + slow	1 run
Rest	18 min	10 min
Out-and-back, 10 laps	yes	–
Marvelmind map	B1 height wrong	correct
Use	characterisation	validation

E. Test campaigns

[TODO: Two campaigns at the same robot poses (Table I). Campaign 1 is the main test programme; it was recorded with an error in the Marvelmind map (the height of beacon 1, 2.6 m, was not applied and the map used 0 m), found afterwards from the height reported for the mobile beacon. Campaign 2 repeats the rotations, the triangle and the rest with the correct map: it is the reference for the correction of campaign 1, and the two campaigns register each other (Section IV-B).]

IV. METHOD

A. Correction of the beacon height error

[TODO: Each Marvelmind position comes from three beacons. From the recorded position and the wrong map, the distances to each triple are rebuilt; the position is recomputed with the corrected map for every triple; the triple whose height matches the known hedgehog height is kept, with a continuity term between samples (dynamic programming). The reference is not used. Checks: the four distances of one sample shown by the Marvelmind software fit the corrected map within 4 mm; the corrected campaign 1 is compared with campaign 2 (Section V-A).]

B. Cross-campaign registration and delay

[TODO: 2D rotation, translation and lever arm between the OptiTrack and Marvelmind frames, robust least squares on the stop medians [7]. No recording is evaluated with parameters fitted on itself: campaign 1 is evaluated with the registration fitted on campaign 2; campaign 2 with the one fitted on the campaign 1 stops with the same hedgehog mounting; the lever arm of the first mounting is fitted on the other rotation run. Delay: median of the per-recording estimates, $0.105(36)$ s.]

C. Monte Carlo of beacon height errors

[TODO: With the true beacon geometry and the true hedgehog trajectories (C1 slow triangle and out-and-back, 3009 samples), each beacon height is offset by ± 0.05 to -2.6 m; the map the Marvelmind software would build from the true beacon distances with that height is recomputed; ranges get 3 mm noise; the solver uses a random triple (pessimistic). Outputs: position error after rigid registration, error of the reported hedgehog height, and the result of the correction.]

D. Data selection and metrics

[TODO: Stops: OptiTrack speed below 2 cm/s and rotation below $3\,^\circ$/s for at least 10 s; frames with marker error above 3 mm discarded. Metrics: 2D error, median, RMS, p95, CDF, per-stop offset and jitter, Allan deviation [6].]

E. Uncertainty budget and Monte Carlo propagation

[TODO: Table II [5]. The intervals of Table III come from a Monte Carlo propagation [8]: 300 draws of the registration (stop bootstrap), the delay ($\mathcal{N}(0.105, 0.036)$ s),

TABLE II
UNCERTAINTY BUDGET OF THE REFERENCE POSITION (2D, STANDARD UNCERTAINTIES).

Contribution	u (cm)
OptiTrack position (mean marker error)	0.11
Registration: rotation, translation, lever arm	3.1–5.4
of which lever arm (per axis)	1.8–2.9
Delay, $0.105(36)$ s at $0.3\,\mathrm{m/s}$	1.1
Synchronisation jitter at $0.3\,\mathrm{m/s}$	0.15
Combined, at rest / moving	5.4 / 5.5
Expanded ($k = 2$), at rest / moving	10.8 / 11.0
Temperature $\pm 3\,^{\circ}\mathrm{C}$ (on Marvelmind)	1.8

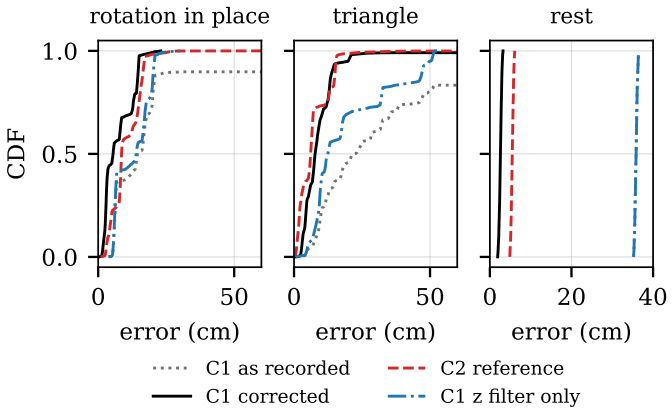


Fig. 2. Error distribution per test type: campaign 1 as recorded (dotted), with the height filter only (dash-dot), after the correction (solid), and campaign 2 with the correct configuration (dashed).

OptiTrack noise and a resampling of the stops. The OptiTrack instrument is not the limit; the registration is, mainly through the lever arm (bootstrap over the stops). It is a common-mode term: differences between campaigns and the effect of the configuration error are well resolved, absolute offsets of a few cm at rest are not. Temperature acts on Marvelmind, not on the reference.]

V. RESULTS

A. Effect of the configuration error and its correction

[TODO: Table III, Fig. 2: the configuration error multiplies the errors by 3–10. Discarding the samples with a wrong height removes the worst outliers but keeps the bias (rest 35.8 cm) and only 55–88 % of the samples. After the correction (81–100 % of the samples kept) campaign 1 agrees with campaign 2 within the resolution of the comparison: median differences -4.0 and 2.0 cm, moving RMS 11.0 vs 11.1 cm, all with 95 % Monte Carlo intervals that include zero. One stop out of 23 failed the correction (167 cm). Sensitivity: median error stable within 1 cm for an assumed hedgehog height ± 5 cm, height tolerance 7–15 cm, with or without the continuity term.]

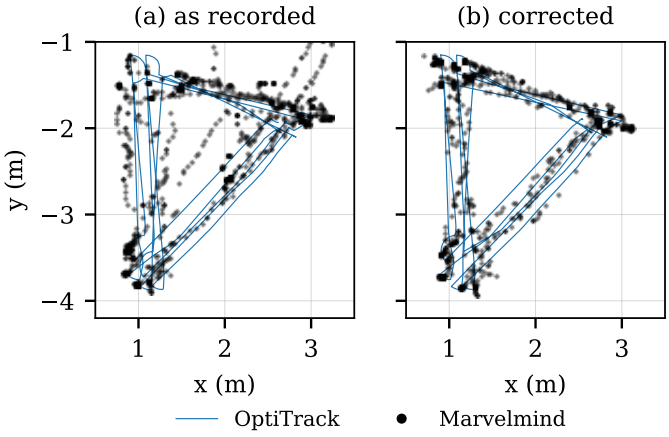


Fig. 3. Campaign 1, slow triangle: Marvelmind positions (dots) against the OptiTrack reference (line), (a) as recorded with the wrong height of beacon 1, (b) after the correction. [TODO: optional]

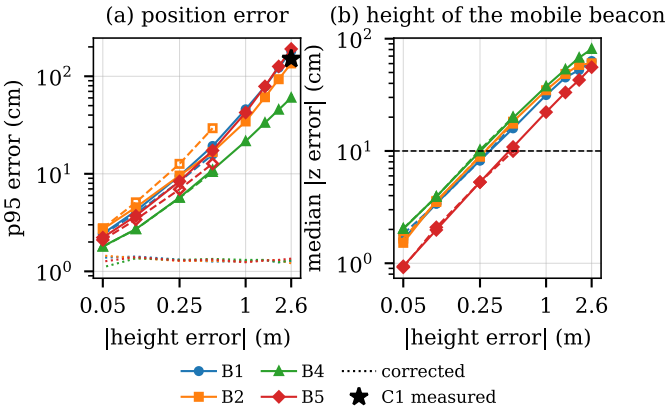


Fig. 4. Monte Carlo of a wrong beacon height, for each beacon. (a) p95 position error against the height error (filled: beacon set too low in the map, open: too high; dotted: after the correction; star: campaign 1 measured). (b) Median error of the height reported for the mobile beacon; dashed: 10 cm.

B. Tolerance to beacon height errors (simulation)

[TODO: Fig. 4. Check: B1 at -2.6 m gives p95 181 cm and 58 % wrong 3D solutions, same order as campaign 1 (p95 51–151 cm, 4–49 %). Height errors up to ± 0.1 m add at most about 5 cm (p95) and hardly show in z; between 0.1 and 0.5 m the position error is already 5–30 cm but only part of the samples show a height error above 10 cm; above 0.5 m the height check reveals it almost always. The correction restores the no-error level in every simulated case (p95 1.2–1.5 cm). Advice: measure beacon heights within ± 5 cm and check the hedgehog height before recording.]

C. At rest

[TODO: Precision vs accuracy (Fig. 5): same noise in both campaigns, jitter 4 mm, Allan deviation 1.6 mm at 1 s and 0.2 mm at 60 s; offset 2.6 and 5.5 cm, within the registration uncertainty. Repeated visits (Fig. 6a): offset per vertex 2–10 cm, spread between visits 1–9 cm, jitter within a stop 1.4 mm. Heading (Fig. 6b): the same pattern in both

TABLE III

MARVELMIND ERROR (CM): CAMPAIGN 1 AS RECORDED (C1 REC.), WITH THE HEIGHT FILTER ONLY (C1 FILT.), CORRECTED (C1 CORR.), AND CAMPAIGN 2 (C2). STOPS: MEDIAN / P95 OF THE PER-STOP OFFSET; MOVING: RMS / P95; ALL: MEDIAN / P95 OF ALL SAMPLES. LAST COLUMN: DIFFERENCE C1 CORR. − C2 WITH THE 95 % MONTE CARLO INTERVAL.

Test	C1 rec.	C1 filt.	C1 corr.	C2	C1 corr. − C2
Rotation, stops	16.6 / 85.8	–	4.3 / 14.0	8.4 / 16.1	−4.0 [−12.8, 2.4]
Triangle, stops	20.7 / 126.7	–	7.8 / 14.2	5.8 / 14.2	2.0 [−3.9, 10.6]
Triangle, moving	65.6 / 141.5	20.4 (RMS)	11.0 / 21.1	11.1 / 23.4	−0.1 [−2.4, 3.7]
Triangle, all	–	12.4 / 50.4	7.7 / 19.9	–	–
Rest	35.9	35.8	2.6	5.5	−2.9 [−7.4, 7.3]
Kept samples	100 %	55–100 %	81–100 %	100 %	

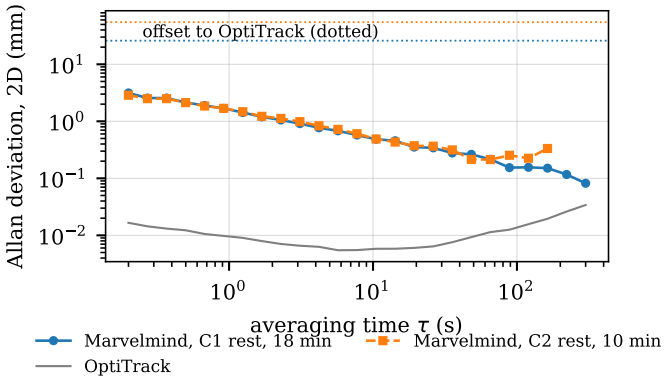


Fig. 5. Allan deviation of the Marvelmind position at rest in the two campaigns, and of OptiTrack. The dotted lines give the offset of each recording from the reference.

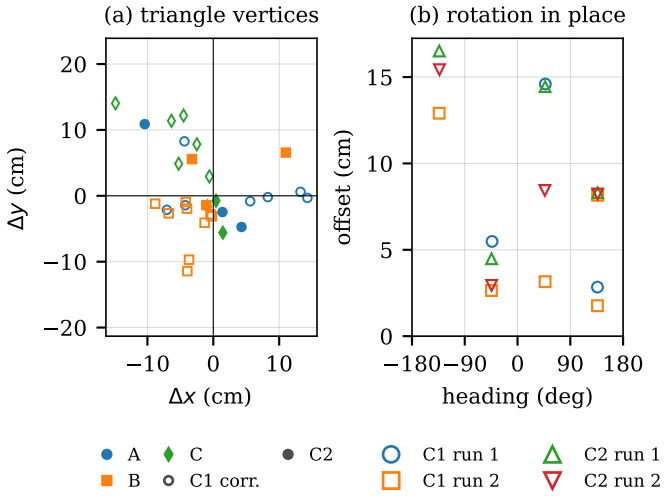


Fig. 6. Offset at rest. (a) Offset at each stop on the three triangle vertices, campaign 1 corrected (open) and campaign 2 (filled). (b) Offset against the robot heading for the rotation in place, two runs per campaign.

campaigns (−44°: 2.6–5.5 cm; −134°: 12.9–16.5 cm), so it belongs to the system. Cause: the beacon left out of the position solution depends strongly on the heading (e.g. B1 in 62 % of the samples with heading 90–180°, 1 % with −90–90°) and lies preferentially on one side/rear of the robot, consistent with shading by the robot body and mast (the triple is inferred by the correction, campaign 1).]

D. While moving

[TODO: RMS 11.0 and 11.1 cm, p95 21–23 cm in the two campaigns (Table III); no clear dependence on the speed up to 0.6 m/s (Fig. 7); delay 0.105(36) s. Temporal filters gained at most 1 cm.]

E. Outside the OptiTrack area

[TODO: Campaign 1 out-and-back: error before leaving 5.4 cm, after re-entry 4.4 cm (median, 10 re-entries); spread at the outside points 6–10 cm, an upper bound (Fig. 8).]

VI. DISCUSSION AND CONCLUSION

[TODO: Lessons for users: check that the mobile beacon reports its real height before recording; log the raw distances so that configuration errors can be corrected exactly. Limits: one room, 2D only, joystick paths, small OptiTrack area in a corner of the beacon area, temperature not measured.]

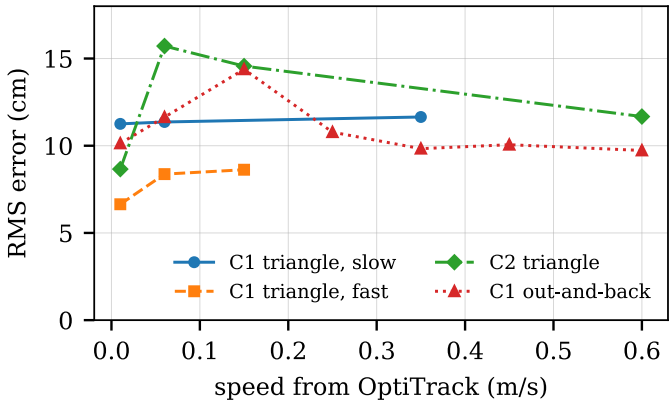


Fig. 7. RMS error of Marvelmind as a function of the speed measured by OptiTrack. [TODO: optional]

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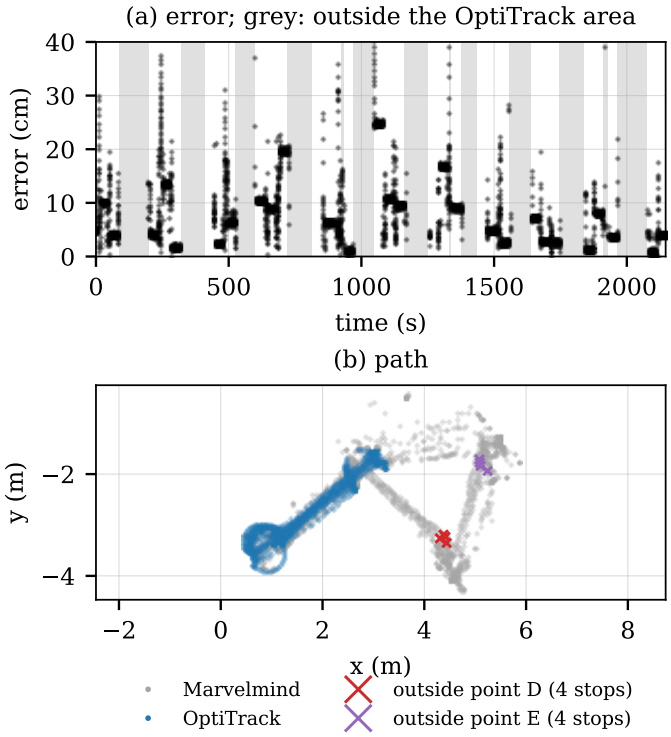


Fig. 8. Out-and-back test. (a) Marvelmind error over time; grey: outside the OptiTrack area. (b) Paths and Marvelmind positions at the outside stops.

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